

SEDAR 20: South Atlantic Croaker and Menhaden Review

South Atlantic Croaker

Short-Term Research Priorities (for next benchmark assessment, in order of importance)

1. Continue fisheries-independent surveys throughout the species range and subsample for individual weights and ages, particularly in the southern range.
2. Encourage fishery-dependent biological sampling of Atlantic croaker from the southern region. Collect age samples from the recreational fishery when the length distribution of the recreational fishery samples is not adequately represented by the fisheries from which the age-length keys are developed.
3. Maintain SEAMAP funding.
4. Increased observer coverage for commercial discards.
5. Hybrid random sampling of commercial catch: sample catch for ageing at random, and mark those samples as selected randomly, then supplement underrepresented length bins with additional samples—this will avoid the necessity of weighting length-at-age estimates by the fisheries length frequencies.
6. Conduct studies of discard mortality for recreational and commercial fisheries.
7. Conduct study on fecundity in the south Atlantic and continue to develop estimates of length-at-maturity and year-round reproductive dynamics.
8. Investigate environmental covariates in stock assessment models.
9. Historical summaries of landings data from NOAA indicate landings are available at a finer scale (e.g., landings by water body, month) for the earliest years than are currently reported. We encourage efforts to recover these data and make them available for stock assessments.
10. Re-examine historical ichthyoplankton studies of the Chesapeake Bay for an indication of the magnitude of estuarine spawning.

Long-Term Research Priorities (in order of importance)

1. Collect data on fishing attributes necessary to develop gear-type-specific fishing effort estimates.
2. Develop and implement sampling programs for state-specific commercial scrap fisheries in order to monitor the relative importance of Atlantic croaker in the scrap landings.
3. Develop a coast-wide tagging program for Atlantic croaker to evaluate migration and movement and continue any coast-wide studies (e.g., genetics, otolith microchemistry) designed to improve understanding of stock definition.
4. Examine socioeconomic aspects of the fishery.

Atlantic Menhaden

Research recommendations are broken down into two categories: data and modeling. While all recommendations are high priority, the first recommendation is the highest priority. Each category is further broken down into recommendations that can be completed in the short term and recommendations that will require long term commitment.

Annual Data Collection

Long term:

1. [Highest Priority] Develop a coastwide fishery independent index of adult abundance at age to replace or augment the existing Potomac River pound net index in the model.
2. Work with industry to collect age structure data outside the range of the fishery.

Short term:

1. Continue current level of sampling from bait fisheries, particularly in the mid-Atlantic and New England, and continue recovery of historical tagging data from paper data sheets.
2. Request annual samples of menhaden from the PRFC pound net fishery to better characterize age and size structure of catch.

Assessment Methodology

Long term:

1. Develop multispecies statistical catch-at-age model to estimate menhaden natural mortality at age.
2. Develop spatially-explicit model, once sufficient age-specific data on movement rates of menhaden is available.

Short term:

1. Re-evaluate menhaden natural mortality-at-age and population response to changing predator populations by updating and augmenting the MSVPA (e.g. add additional predator, prey, and diet data when available).
2. Incorporate maturity at age variability in the assessment model.

Future Research

Evaluate productivity of different estuaries (e.g., replicate similar methodology to Ahrenholz et al. 1987).

2. Collect age-specific data on movement rates of menhaden to develop regional abundance trends.
3. Determine selectivity of PRFC pound nets.
4. Update information on maturity, fecundity, spatial and temporal patterns of spawning, and larval survivorship.
5. Investigate the effects of global climate change on distribution, movement, and behavior of menhaden.